



CPEG 418 · COMPUTER VISION PROJECT PROPOSAL

PhotoMatch

Local Face Recognition for Instant Event Photo Retrieval

Department of Computer Engineering, American University of Kuwait

A photography business outgrowing manual delivery

The client

Golden Camera, a freelance photography business shooting weddings, corporate events, school graduations, and university graduations (even our university) often thousands of photos per event.

The workflow today

Once an event is shot and edited, all photos are dumped into a shared folder. Students and guests then scroll through the entire set by eye to find themselves.

10,000+

photos per graduation event

5–6 hrs

spent per student searching manually

Peak season

many students need photos at once

Manual photo retrieval is the bottleneck

Editing is no longer the slowest step in the business finding the right photos for the right person is. Every event ends with hours of unpaid, error-prone manual sorting that delays delivery and caps how many events the business can serve.



Slow

5–6 hours of manual searching per student across 10,000+ unsorted photos.



Error-prone

Easy to miss photos or hand a student the wrong set entirely.



Unscalable

Bottleneck grows worse as event count and photo volume increase.

What the system needs to achieve



Automate retrieval

Let a student find their own photos in seconds by uploading a single selfie.



Keep it local

Run entirely on the business's own hardware — no cloud upload, strong privacy story.



Scope by event

Tag every photo with an event ID so searches stay fast and accurate.



Handle real conditions

Work reliably across angles, lighting, group shots, and partial occlusion (caps/gowns).



Reduce false negatives

Surface “possible matches” separately so a real photo is never silently missed.



Grow with the business

Design the index to scale from 10,000 to 100,000+ photos across events.

Where the project draws its boundaries

In Scope

- Face detection & recognition pipeline on local hardware
- Event-scoped photo indexing (per event_id)
- Selfie-based student search & retrieval
- Manual trigger to index only fully-edited events
- Confidence-tiered results (confident vs. possible matches)
- Basic evaluation: precision/recall, FAR/FRR, query time

Out of Scope (v1)

- Cloud sync or remote/multi-branch access
- Payment processing & watermark release automation
- Mobile app / desktop/local kiosk workflow only
- Full liveness/anti-spoofing hardening (basic check only)
- Automatic identity tagging without any student selfie

Building on established CV building blocks

Face detection

Haar cascades / MTCNN / RetinaFace

Locating faces within a full event photo before recognition.

Face embeddings

FaceNet / ArcFace-style embedding models

Mapping each detected face to a compact vector for comparison.

Similarity search

Cosine similarity, nearest-neighbor indexing (e.g., FAISS)

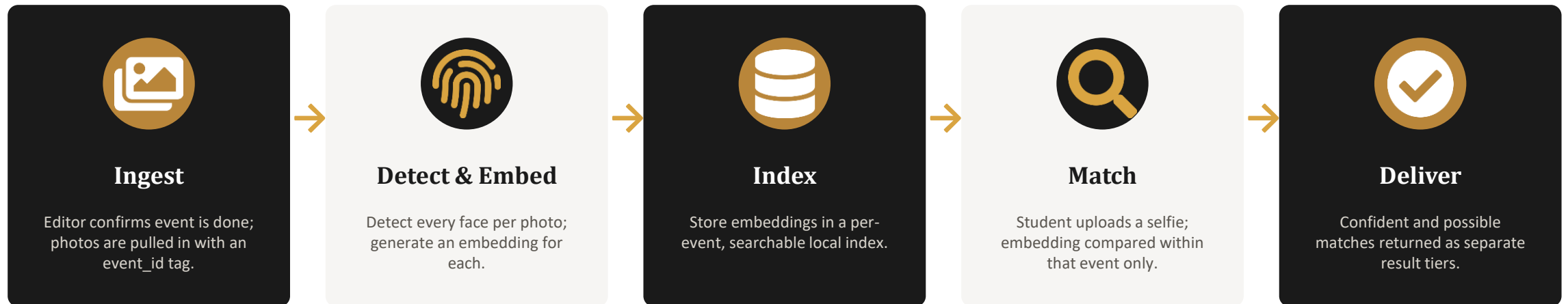
Matching a student's selfie embedding to stored event photo embeddings.

Classical CV toolkit

OpenCV (transforms, morphology, segmentation)

Preprocessing: normalization, denoising, and image cleanup pre-detection.

From event folder to student selfie match



Manual/hybrid trigger keeps unfinished or soon-to-be-replaced photos out of the index.

The stack behind PhotoMatch



Python + OpenCV

Core image I/O, preprocessing, and pipeline glue — built on CPEG 418 lab work.



Face detection / embedding model

Pretrained local model (e.g., MTCNN + FaceNet/ArcFace-style embeddings).



Local vector index

FAISS or equivalent for fast per-event nearest-neighbor similarity search.



Local storage only

All photos, embeddings, and search run on-premise no external upload.



matplotlib

Visualizing detections, matches, and evaluation results for the report.



Simple local UI

Lightweight desktop/kiosk interface for selfie upload and results review.

Real event photos, not clean benchmarks

Source

Real photos from the business's own weddings, events, and school/university graduations the same set staff currently sort by hand.

Why real data matters

Clean benchmark datasets don't capture this business's real challenges: varied angles, mixed lighting, group shots, motion blur, and caps/gowns partially covering faces. Testing on an early real sample keeps the design honest.

Sensitivity

Photos contain biometric data of real people, often minors handled under the consent and privacy plan on the following slides.

10,000+

photos per graduation event today

50,000–100,000+

target scale as the business and event count grow

Multiple event types

weddings · corporate events · school & university graduations

Cleaning images before detection runs

01

Color & geometry correction

Normalize color balance and correct geometric distortion across mixed-lighting event shots.

02

Intensity transformation

Adjust brightness/contrast so faces are detectable in both bright outdoor and dim indoor shots.

03

Convolution-based filtering

Denoise and sharpen images affected by motion blur or low light before face detection.

04

Morphological operations

Clean up detection masks remove noise, fill small gaps around detected face regions.

05

Segmentation

Isolate face regions from cluttered group-photo backgrounds prior to embedding extraction.

Detection, embedding, and matching logic



Face detector

Locates every face in a photo, including partially obscured faces in group / cap-and-gown shots.



Embedding model

Converts each detected face into a fixed-length vector capturing identity, robust to angle/lighting.



Multi-face handling

Each face in a group photo gets its own embedding; the photo matches if any face matches.



Threshold-tuned matching

Cosine similarity vs. selfie embedding, with confident and possible-match tiers.



Auto-clustering (stretch)

Group similar faces per event into identity clusters staff can quickly label.

Proving the system works not just demoing it



Precision / Recall @ top-k

How many of the top retrieved photos are correct, and how many correct photos are found.



False Acceptance Rate (FAR)

How often the system wrongly matches a photo to the wrong person.



False Rejection Rate (FRR)

How often the system misses a photo that actually belongs to the student.



Avg. query time per selfie

End-to-end latency from selfie upload to returned results.



Time saved vs. baseline

Measured against the current 5–6 hour manual search per student.

What success looks like



Minutes-to-seconds retrieval

Student photo search drops from a 5–6 hour manual task to a seconds-long selfie search.



Privacy-first deployment

Fully local pipeline with a defined consent and data-retention policy for biometric data.



Measured accuracy

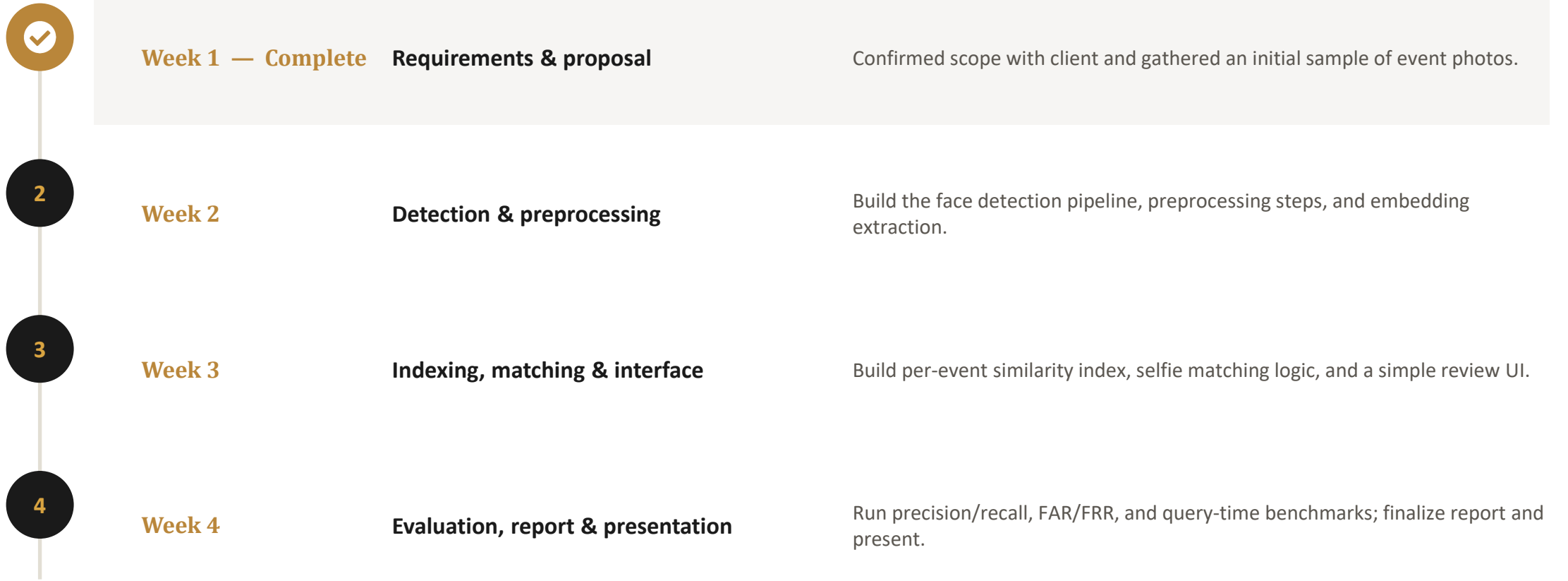
Documented precision/recall, FAR, and FRR on the business's real event photos.



A tool staff will actually use

Simple enough for non-technical workers to trigger indexing and review matches.

4-week summer course plan



Minimal cost built on existing infrastructure



No dedicated budget required

The system runs entirely on the shop's and workers' existing computers. Software components (OpenCV, pretrained detection/embedding models, vector search library) are open-source with no licensing cost.

Hardware	Existing shop / workstation computers no new purchase planned
Software & models	Open-source (OpenCV, detection/embedding models, local vector index)
Optional future spend	Faster storage or a dedicated indexing machine if volume grows well past 100,000 photos

Who's building PhotoMatch



Mahmood Tendail

ID 66597 · Project Lead & Face Detection

Owns the detection/embedding pipeline; coordinates integration.



Ali Mezher

ID 61392 · Preprocessing Engineer

Handles color, intensity, and convolution-based cleanup.



Malek AlKashat

ID 61513 · Indexing & Search

Builds the per-event embedding index and similarity search.



Hazem Ahmed

ID 70493 · Matching Logic

Tunes thresholds for confidence-tiered match results.



Nourallah Mourad

ID 59762 · Interface & Clustering

Builds the selfie UI and auto-clustering feature.



Ahmad AlAli

ID 64024 · Evaluation & Report

Runs evaluation metrics and writes the report.

Work is divided across the pipeline so every member owns one stage end-to-end all six collaborate closely at integration points.

Grounding the approach

- Viola, P. & Jones, M. Rapid Object Detection using a Boosted Cascade of Simple Features (foundational face detection).
- Schroff, F., Kalenichenko, D., & Philbin, J. FaceNet: A Unified Embedding for Face Recognition and Clustering.
- Deng, J. et al. ArcFace: Additive Angular Margin Loss for Deep Face Recognition.
- Zhang, K. et al. MTCNN: Joint Face Detection and Alignment using Multi-task Cascaded CNNs.
- OpenCV documentation image processing, transforms, and morphology reference used throughout CPEG 418.
- Johnson, J., Douze, M., & Jégou, H. FAISS: a library for efficient similarity search.



Thank you

Questions & discussion welcome.
